Comparative RLC and Circuit-Level Analysis of Copper vs. CNT Bundle Interconnects for Nanoscale VLSI

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Submitted by

Nishan Kumar Dutta	Roll No: 1181700239
Anupam Bhowmik	Roll No: 1181700241
Kunal Rabidas	Roll No: 1181700253
Angkita Konwar	Roll No: 1181700263

Under the guidance of
Dr. Kavicharan Mummaneni
Assistant Professor
Department of Electronics and Communication Engineering
National Institute of Technology Silchar

Department of Electronics and Communication Engineering
NATIONAL INSTITUTE OF TECHNOLOGY SILCHAR
Assam

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Declaration of Authorship

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Nishan Kumar Dutta	Anupam Bhowmik	Kunal Rabidas	Angkita Konwar
Roll No.:	**Roll No.:**	**Roll No.:**	**Roll No.:**
1181700239	1181700241	1181700253	1181700263

Department of Electronics and Communication Engineering

National Institute of Technology Silchar, Assam

Certificate

This is to certify that the report entitled

Comparative RLC and Circuit-Level Analysis of Copper vs. CNT Bundle Interconnects for Nanoscale VLSI

submitted by

Nishan Kumar Dutta

(Roll No.: 1181700239)

Anupam Bhowmik

(Roll No.: 1181700241)

Kunal Rabidas

(Roll No.: 1181700253)

Angkita Konwar

(Roll No.: 1181700263)

to the Department of Electronics and Communication Engineering, National Institute of Technology Silchar in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology** in **Electronics and Communication Engineering** is a bonafide record of the work carried out under my supervision. It is further certified that the candidates have complied with all formal requirements.

Dr. Kavicharan Mummaneni

Supervisor,

Assistant Professor, Dept. of ECE,

National Institute of Technology Silchar

Silchar, 788010

Official Seal

Department of ECE

National Institute of Technology Silchar

Silchar, 788010

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Nishan Kumar Dutta	Anupam Bhowmik	Kunal Rabidas	Angkita Konwar
Roll No.:	**Roll No.:**	**Roll No.:**	**Roll No.:**
1181700239	1181700241	1181700253	1181700263

Department of Electronics and Communication Engineering

National Institute of Technology Silchar, Assam

Abstract

The relentless scaling of VLSI technology has pushed copper (Cu) interconnects toward a fundamental resistivity bottleneck, as surface and grain-boundary scattering effects dominate at sub-14nm dimensions, threatening the reliability and speed of future integrated circuits. This project presents a comparative RLC and circuit-level analysis of copper and single-walled carbon nanotube (SWCNT) bundle interconnects across three technology nodes (45nm, 14nm, and 7nm) and three interconnect lengths (100, 500, and 1000 μm). Resistance, capacitance, and inductance were derived analytically for each material: Cu resistivity combines the Fuchs-Sondheimer surface scattering and Mayadas-Shatzkes grain-boundary scattering models, capacitance follows the Sakurai-Tamaru formulation, and self-inductance uses the Neumann rectangular-wire expression. The CNT bundle model accounts for quantum and contact resistance, phonon-limited scattering, kinetic and magnetic inductance, and combined quantum-electrostatic capacitance, drawn from recent SWCNT interconnect literature. All formulas were verified and cross-checked for dimensional and physical consistency before use. The extracted RLC values were embedded into circuit netlists for a CMOS inverter chain and a 4-bit ripple-carry adder, enabling transient simulation of propagation delay and signal integrity under Cu versus CNT interconnect loading. Results show Cu resistivity rising sharply below 14nm, while CNT bundles exhibit higher resistance at short lengths due to quantum and contact resistance dominance, but scale more favorably with increasing length and technology node. The study underscores copper's growing resistivity bottleneck at advanced nodes and positions CNT bundles as a promising, contact-resistance-limited alternative for future VLSI interconnects.

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Chapter 1

Introduction

1.1 General Introduction

The continuous scaling of complementary metal-oxide-semiconductor (CMOS) technology, as projected historically by the International Technology Roadmap for Semiconductors (ITRS) and, more recently, the International Roadmap for Devices and Systems (IRDS) [1]–[3], has reduced on-chip wire dimensions into the deep sub-20 nm regime. While transistor switching speed has continued to improve with scaling, the resistance–capacitance (RC) delay of the on-chip metal interconnects that route signals between transistors has become the dominant bottleneck in modern very-large-scale-integration (VLSI) circuits.

Copper (Cu), adopted industry-wide in the late 1990s to replace aluminium because of its lower bulk resistivity and superior electromigration resistance, is itself now a limiting factor at advanced nodes. As wire width and height shrink toward, and below, the electron mean free path of bulk Cu ($\approx$39 nm at 21°C), conduction electrons scatter increasingly at the wire surfaces and at grain boundaries within the polycrystalline film. In addition, a thin tantalum-based diffusion barrier (TaN/Ta), required to prevent Cu from diffusing into the surrounding dielectric, must be carved out of the wire's already-shrinking cross-section, consuming close to 29% of the wire width at the 7 nm node. Together, these effects can raise Cu's effective resistivity by an order of magnitude or more relative to its bulk value, degrading interconnect delay, increasing power dissipation, and worsening electromigration reliability.

1.2 Motivation

This deterioration in Cu performance at scaled nodes has motivated a substantial body of research into alternative interconnect conductors, including ruthenium, cobalt, graphene, and carbon nanotubes (CNTs). Single-walled carbon nanotubes (SWCNTs), in particular, are attractive because a metallic SWCNT can carry current densities several orders of magnitude higher than Cu, exhibits a room-temperature electron mean free path on the order of a micrometre, far longer than the physical dimensions of a scaled interconnect, and offers high thermal conductivity and mechanical robustness. When arranged as densely packed bundles running through a Cu-style trench, SWCNT bundles have been proposed as a potential replacement for local, intermediate, and even global on-chip wiring.

However, an individual SWCNT also carries substantial quantum (contact) resistance, and a real bundle grown by present-day synthesis techniques contains a mixture of metallic and semiconducting tubes, so that only a fraction of the bundle actually conducts. Quantifying whether, and under what conditions, CNT bundles can outperform Cu at a given technology node and wire length is therefore an open and practically important question, and is the central motivation for this project.

1.3 Problems

The comparative evaluation of Cu and CNT bundle interconnects presents several challenges:

- **Non-Linear Resistivity Scaling in Cu:** Surface and grain-boundary scattering effects cause Cu resistivity to rise sharply and non-linearly below 14 nm, requiring physically accurate rather than bulk-value resistivity models.
- **Barrier Layer Erosion:** The TaN/Ta diffusion barrier consumes a growing fraction of the already-shrinking wire cross-section at advanced nodes, further degrading effective Cu conductivity.
- **Quantum and Contact Resistance in CNT Bundles:** A single SWCNT carries a fixed quantum resistance of $h/4e^2 \approx 6.45\ \mathrm{k}\Omega$ regardless of length, and additional imperfect metal–CNT contact resistance, both of which dominate at short interconnect lengths.

- **Metallic Tube Yield Uncertainty:** Present-day unsorted CNT synthesis produces a mixture of metallic and semiconducting tubes, so only a fraction of a bundle's tubes conduct, introducing significant modelling uncertainty.
- **Consistent Multi-Node, Multi-Length Comparison:** Existing literature rarely applies a single self-consistent RLC framework across a wide range of technology nodes and interconnect lengths simultaneously.
- **Circuit-Level Validation:** RLC parameter comparisons alone do not capture real circuit behaviour; propagation delay and signal integrity must be validated through transient simulation.

1.4 Objectives of the Work

The primary objective of this work is to quantitatively compare copper and single-walled carbon nanotube bundle interconnects across multiple technology nodes and interconnect lengths using a consistent analytical and circuit-simulation framework. The specific objectives are:

- To model the resistance, inductance, and capacitance (RLC) of Cu interconnects at the 45 nm, 14 nm, and 7 nm technology nodes using the combined Fuchs-Sondheimer/Mayadas-Shatzkes resistivity model, the Sakurai-Tamaru capacitance model, and the Neumann self-inductance formula.
- To model the RLC parameters of SWCNT bundle interconnects at the same nodes using a phonon-scattering-limited resistance model together with quantum, magnetic, and electrostatic inductance/capacitance contributions.
- To evaluate both interconnect types at three representative wire lengths, 100 µm, 500 µm, and 1000 µm, corresponding to local, intermediate, and global interconnect tiers.
- To compare the resulting R, L, and C values across all node-length combinations and identify the technology nodes and lengths at which SWCNT bundles most closely approach, or could exceed, Cu's electrical performance.
- To embed the extracted RLC values into circuit-level netlists (CMOS inverter chain and 4-bit ripple-carry adder) and validate the analytical trends through transient propagation-delay simulation.

- To document all modelling assumptions transparently so the analysis can be extended to delay (Elmore), power-delay product (PDP), and energy-delay product (EDP) evaluation in subsequent work.

1.5 Scope and Assumptions

The geometric parameters used for each technology node, wire width, aspect ratio, spacing, dielectric constant, and barrier thickness, are taken from the ITRS 2009 interconnect chapter (45 nm), ITRS 2.0 (2015) (14 nm), and IRDS 2022 (7 nm) [1]–[3].

The SWCNT bundle is assumed to be produced by an unsorted synthesis process, giving a metallic tube fraction $F_m = 1/3$, **with a representative metal–CNT contact resistance of 10 kΩ per tube**. Both parameters are treated as adjustable assumptions rather than measured values, and their influence on the results can be examined through sensitivity analysis in future work. All calculations correspond to a single, isolated signal line at room temperature (300 K); coupling to adjacent lines beyond the spacing-dependent fringe-capacitance term already built into the Cu model is outside the present scope.

1.6 Organization of the Report

This report is organized into six chapters. Chapter 2 reviews existing literature on Cu resistivity scaling, interconnect capacitance modelling, CNT interconnect physics, and comparative Cu–CNT studies, concluding with the identified research gap. Chapter 3 presents the methodology, including the analytical RLC formulations for both materials and the circuit-level simulation setup. Chapter 4 presents the results and discussion, comparing R, L, and C trends across nodes and lengths, alongside circuit-level propagation delay behaviour. Chapter 5 concludes the report with key findings, limitations, and future scope. Chapter 6 lists the references used throughout this study.

Chapter 2

Literature Review

2.1 Copper Interconnect Scaling and the Resistivity Problem

The interconnect scaling trends and geometric projections used throughout this project are drawn from three successive editions of the international semiconductor roadmap: the ITRS 2009 interconnect chapter, ITRS 2.0 (2015), and IRDS 2022 [1]–[3], which together track the transition from planar 45 nm wiring to barrier-dominated sub-10 nm wiring.

The physical origin of Cu's resistivity increase at these scaled dimensions was established by the classical theories of surface scattering, developed by Fuchs (1938) and Sondheimer (1952), and of grain-boundary scattering, developed by Mayadas and Shatzkes (1970); these remain the basis of essentially all subsequent Cu interconnect resistivity models. Smith et al. [4] revisited both theories for wide, sub-14 nm Cu lines and showed experimentally that the simplified linear form of the Fuchs–Sondheimer expression, commonly used in older VLSI papers, breaks down once wire dimensions approach the Cu mean free path ($\approx$39 nm), and proposed use of the full Fuchs–Sondheimer integral together with a negative specularity parameter ($p \approx -0.2$) for accurate prediction below 14 nm.

Camacho and Oliva [5] independently combined the Fuchs–Sondheimer integral and Mayadas–Shatzkes grain-boundary term into a single unified resistivity expression while surveying epitaxial metals as potential Cu replacements. Saligram et al. [6] extended this resistivity-modelling approach specifically to the 7 nm node, comparing scaled Cu against ruthenium and confirming that, once the TaN/Ta diffusion barrier is accounted for, effective Cu resistivity can exceed twenty times its bulk value. These three works collectively form the basis of the Cu resistivity model adopted in this project.

2.2 Capacitance Modelling of On-Chip Interconnects

Interconnect capacitance is dominated by both the parallel-plate component between a wire and its reference plane and the fringing-field component to neighbouring structures. Sakurai and Tamaru [7] derived closed-form two- and three-dimensional capacitance expressions for a single wire over a ground plane that remain widely used for their computational simplicity. Later VLSI interconnect papers, including the comparative Cu–CNT study of Li et al. [8], adopted an extended three-term variant of this formula with ITRS-fitted exponents to better capture fringing capacitance to adjacent lines as a function of wire spacing; this extended form, rather than the original two-term 1983 expression, is the version used for the Cu capacitance calculations in this project, consistent with its established use in the comparative-interconnect literature.

2.3 Carbon Nanotubes as an Interconnect Material

A single-walled carbon nanotube can be either metallic or semiconducting depending on its chirality, with roughly one-third of tubes produced by conventional (unsorted) synthesis being metallic. A metallic SWCNT supports four conducting channels and exhibits ballistic transport over lengths shorter than its electron mean free path, but is limited by a fixed quantum (contact) resistance of $h/4e^2 \approx 6.45$ kΩ regardless of length, plus an additional imperfect metal–CNT contact resistance that can range from roughly 3.2 kΩ for best-case palladium contacts to 100 kΩ for poor contacts.

Park et al. [9] experimentally characterised electron–phonon scattering in metallic SWCNTs and reported acoustic- and optical-phonon mean free paths of approximately 400 nm (scaling as $1/T$) and 56 nm respectively at room temperature, values that combine, via Matthiessen's rule, into an effective scattering mean free path used to model length-dependent CNT resistance. Srivastava et al. [10] provided the first complete three-dimensional treatment of SWCNT bundle capacitance, introducing the quantum capacitance, arising from the finite density of states at the Fermi level in a 1-D conductor, alongside the classical electrostatic capacitance to ground.

Naeemi and Meindl [11] developed a comprehensive performance model for both single- and multi-wall CNT interconnects that, in addition to resistance and capacitance, incorporated the kinetic inductance associated with electron inertia in 1-D conduction channels, a compo-

nent with no analogue in bulk metal wires, and one that this project's results show becomes the dominant inductance term in CNT bundles at scaled nodes. Building on these foundations, Majumder et al. [12] presented a more recent and more complete SWCNT bundle model, combining temperature-dependent acoustic- and optical-phonon scattering into a single effective mean free path and updating the quantum and electrostatic capacitance expressions for ITRS/IRDS node-wise geometries; this model is adopted as the primary CNT model for this project.

2.4 Comparative Cu–CNT Interconnect Studies

Several studies have directly compared Cu and CNT interconnect performance. Naeemi et al. [13] performed one of the earliest such comparisons for gigascale integration and showed that CNT bundles could outperform Cu in delay at longer interconnect lengths, provided a sufficiently high density of metallic tubes and low contact resistance could be achieved. Srivastava and Banerjee [14] similarly analysed CNT interconnect performance for VLSI applications and highlighted the sensitivity of the comparison to bundle density and contact quality.

Li et al. [8] modelled both Cu and CNT interconnects at the 45 nm node using a common RLC framework and reported that CNT bundles became competitive with Cu primarily for longer, global-tier wires. More recently, Parihar and Sharma [15] compared a mixed CNT bundle against Cu at the 32 nm node using the Neumann inductance formulation adopted here, and used the energy-delay product (EDP) as the principal figure of merit, finding that CNT bundles could achieve a lower EDP than Cu for interconnects longer than roughly 200 µm at 22 nm and below, even though this advantage does not always translate into a strict resistance improvement at every node.

Taken together, these studies establish that the Cu–CNT comparison is highly sensitive to three parameters that vary widely across the literature: the metallic tube fraction F_m (from $\approx$1/3 for unsorted synthesis up to nearly 1.0 for sorted or aligned tubes), the metal–CNT contact resistance (3.2–100 kΩ), and the interconnect length under consideration.

Table 2.1: Literature Review survey on Cu Resistivity, Capacitance, and CNT Interconnect Modeling

SR	Paper Title	Author(s)	Year	Methodology	Advantages
1	Evaluation of Fuchs-Sondheimer and Mayadas-Shatzkes models below 14 nm	R. S. Smith et al.	2019	Experimental resistivity fitting, negative specularity	Accurate sub-14nm Cu resistivity prediction
2	Epitaxial metals for interconnects beyond Cu	J. M. Camacho and A. I. Oliva	2020	Unified FS+MS resistivity model	Combines surface and grain-boundary scattering
3	Resistivity Modeling of Scaled Cu and Ru Interconnects	R. Saligram et al.	2022	7nm resistivity modeling with barrier layer	Quantifies TaN/Ta barrier impact
4	Simple formulas for two- and three-dimensional capacitances	T. Sakurai and K. Tamaru	1983	Closed-form capacitance expressions	Computationally simple, widely used
5	Modeling of CNT Interconnects and Comparative Analysis with Cu	H. Li et al.	2006	Common RLC framework, ITRS-fitted capacitance	Extended fringing capacitance model
6	Electron-phonon scattering in metallic SWCNTs	J. Y. Park et al.	2004	Experimental phonon mean free path measurement	Basis for CNT scattering-limited resistance

SR	Paper Title	Author(s)	Year	Methodology	Advantages
7	Carbon Interconnects: Implications for Performance and Power	N. Srivastava et al.	2005	3D SWCNT bundle capacitance treatment	Introduced quantum capacitance term
8	Performance Modeling for Single- and Multiwall CNT Interconnects	A. Naeemi and J. D. Meindl	2008	CNT performance model with kinetic inductance	First to model 1-D kinetic inductance
9	Theoretical Modeling of Adaptive Mixed CNT Bundles	M. K. Majumder et al.	2022	Combined phonon-scattering CNT model	Primary CNT model adopted in this work
10	Performance comparison between CNT and Cu for GSI	A. Naeemi et al.	2005	Early Cu-CNT delay comparison	Identified CNT advantage at long lengths
11	Performance analysis of CNT interconnects for VLSI	N. Srivastava and K. Banerjee	2005	Bundle density/contact sensitivity analysis	Highlighted contact quality dependence
12	Comparative study of Mixed CNT bundle with Cu at 32nm	T. Parihar and A. Sharma	2013	EDP-based comparison, Neumann inductance	EDP figure of merit for CNT vs Cu

2.5 Research Gap and Problem Statement

Most existing comparative studies were published before the experimentally validated sub-14 nm Cu resistivity model of Smith et al. [4] and the updated phonon-scattering-based CNT

bundle model of Majumder et al. [12] became available, and few, if any, apply a single self-consistent RLC framework across the specific 45 nm, 14 nm, and 7 nm node range together with the 100 µm, 500 µm, and 1000 µm length range that spans local, intermediate, and global on-chip wiring tiers.

This project addresses that gap by combining the latest validated Cu and CNT analytical models within one consistent set of ITRS/IRDS-derived geometric assumptions, enabling a direct, up-to-date, node-by-node and length-by-length comparison between the two interconnect technologies, further validated through circuit-level transient simulation.

Chapter 3

Proposed Methodology

3.1 Interconnect Geometry per Technology Node

All electrical parameters depend first on the physical wire dimensions at each technology node. Table 3.1 lists the nominal geometry used throughout this work, drawn from the ITRS 2009 interconnect chapter (45 nm), ITRS 2.0 2015 (14 nm), and IRDS 2022 (7 nm).

Table 3.1: Interconnect geometry parameters by technology node

Parameter	**45 nm**	**14 nm**	**7 nm**	**Source**
Wire Width W (nm)	45	14	7	ITRS 2009 / IRDS 2022
Aspect Ratio (H/W)	1.7	1.9	2.0	ITRS 2009
Wire Height H (nm)	76.5	26.6	14	Derived from AR
Wire Spacing S (nm)	45	14	7	ITRS ($S = W$ assumed)
Dielectric const. k	3.0	2.5	2.3	ITRS 2015 / IRDS 2022
Barrier thickness (nm)	3.5	1.5	1.0	ITRS 2009
Effective Cu width (nm)	38	11	5	$W - 2\times$barrier

Parameter	45 nm	14 nm	7 nm	Source
Grain size D (nm)	45	14	7	$D \approx W$
CNT diameter d (nm)	1.6	1.0	0.8	Naeemi and Meindl
CNT bundle F_m	0.33	0.33	0.33	Unsorted synthesis (1/3)

The Cu wire's effective conducting cross-section accounts for the TaN/Ta diffusion barrier: effective width $W_{eff} = W - 2t_{barrier}$ and effective height $H_{eff} = H - t_{barrier}$. At the 7 nm node the barrier consumes approximately 29% of the nominal wire width, which is included explicitly in all Cu resistance calculations. Figure 3.1 illustrates the resulting cross-sectional geometry for the Cu wire and the SWCNT bundle occupying the same trench.

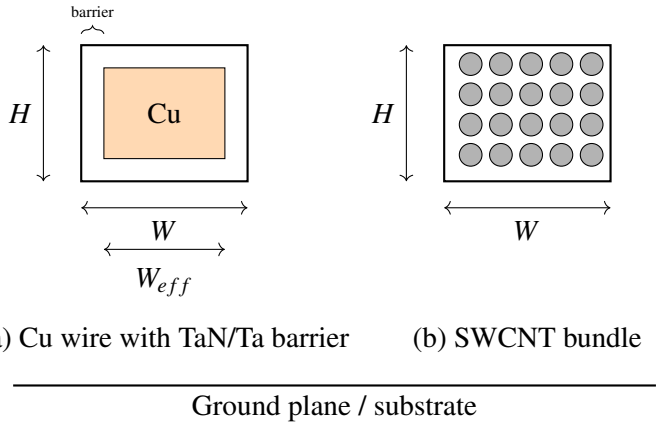

Figure 3.1: Cross-sectional geometry of (a) the copper interconnect showing the TaN/Ta diffusion barrier reducing the effective conducting width, and (b) the SWCNT bundle occupying an equivalent trench, with individual metallic and semiconducting tubes packed at van der Waals spacing $\delta = 0.34$ nm.

3.2 Copper Interconnect RLC Model

3.2.1 Resistance: Combined Fuchs-Sondheimer and Mayadas-Shatzkes Model

Cu resistivity at scaled nodes is modelled by combining the Fuchs-Sondheimer (FS) surface-scattering formulation with the Mayadas-Shatzkes (MS) grain-boundary-scattering formula-

tion, following the experimentally validated sub-14 nm treatment of Smith et al. [4].

The full FS expression for surface scattering is:

$$\rho_{FS} = \rho_0 \times \left[1 - \frac{3\lambda}{2d}(1-p)\int_1^{\infty}\left(\frac{1}{t^3} - \frac{1}{t^5}\right)\frac{1-e^{-kt}}{1-pe^{-kt}}\,dt\right]^{-1} \tag{3.1}$$

where $k = d/\lambda$, $d = \min(W,H)$ is the limiting dimension, $\lambda = 39$ nm is the Cu mean free path at 21°C, and p is the specularity parameter. A specularity of $p = -0.2$ is used for sub-14 nm nodes, consistent with the experimental fit of Smith et al. [4]. For the 45 nm node, where $d \gg \lambda$, the simplified linear approximation

$$\frac{\rho_{FS}}{\rho_0} = 1 + \frac{3\lambda}{8d}(1-p) \tag{3.2}$$

is also valid.

The MS expression for grain-boundary scattering is:

$$\rho_{MS} = \rho_0 \times \left[1 - \frac{3\alpha}{2} + 3\alpha^2 - 3\alpha^3 \ln\left(1 + \frac{1}{\alpha}\right)\right]^{-1}, \qquad \alpha = \frac{\lambda}{D}\cdot\frac{R}{1-R} \tag{3.3}$$

where $D \approx W$ is the grain size and $R = 0.16$–0.17 is the grain-boundary reflection coefficient.

The two mechanisms are combined multiplicatively to give the effective resistivity, following standard practice in the literature:

$$\rho_{eff} = \rho_{FS} \times \frac{\rho_{MS}}{\rho_0}, \qquad R_{Cu} = \frac{\rho_{eff}}{W_{eff} \times H_{eff}} \quad [\Omega/\mathrm{m}] \tag{3.4}$$

Table 3.2 summarises the resulting effective resistivity at each node, showing an increase from ≈3.1 μΩcm at 45 nm to ≈38 μΩcm at 7 nm, over 22× the bulk value of 1.68 μΩcm.

Table 3.2: Cu effective resistivity by technology node

Node	ρ_{bulk} (μΩcm)	**FS correction**	**MS correction**	ρ_{eff} (μΩcm)
45 nm	1.68	∼1.4×	∼1.3×	∼3.1
14 nm	1.68	∼3.2×	∼2.4×	∼12.9
7 nm	1.68	∼5.8×	∼3.9×	∼38.0

3.2.2 Capacitance: Sakurai-Tamaru Formulation

Total Cu interconnect capacitance per unit length combines the parallel-plate and fringing-field contributions:

$$C = \varepsilon_0 \varepsilon_r \left[\frac{W}{H} + 2.04 \left(\frac{S}{S+0.54H} \right)^{1.77} + 1.14 \left(\frac{W}{H} \right)^{0.06} \left(\frac{S}{S+1.51H} \right)^{4.53} \right] \tag{3.5}$$

where $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m and ε_r is the node-dependent dielectric constant from Table 3.1.

This three-term expression is the ITRS-fitted variant of the original Sakurai-Tamaru two-term formula and is adopted here following Li et al. [8], who applied the same extended form with ITRS-fitted exponents for 45 nm-node Cu-CNT comparison. The original 1983 Sakurai-Tamaru expression [7] is a simpler two-term form without spacing dependence; the extended three-term formula used in this work is cited as the Li et al. [8] ITRS-fitted variant rather than the original 1983 paper, consistent with its established use in the comparative-interconnect literature.

3.2.3 Inductance: Neumann Formula

Cu self-inductance is modelled using the Neumann formula for a rectangular cross-section wire:

$$L_{self} = \frac{\mu_0}{2\pi} \left[\ln \left(\frac{2l}{W+H} \right) + 0.5 + 0.2235 \frac{W+H}{l} \right] \quad \text{[H, total]} \tag{3.6}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m and l is the wire length, following the standard form adopted by Parihar and Sharma [15]. Per-unit-length inductance is $L_{per_length} = L_{self}/l$.

3.3 SWCNT Bundle Interconnect Model

3.3.1 Bundle Geometry

The number of CNTs packed into the bundle cross-section under hexagonal packing is:

$$N_{total} = \left\lfloor \frac{W}{d+\delta} \right\rfloor \times \left\lfloor \frac{H}{d+\delta} \right\rfloor \tag{3.7}$$

where d is the CNT diameter and $\delta = 0.34$ nm is the van der Waals inter-tube spacing. The number of metallic tubes is $N_m = \text{round}(N_{total} \times F_m)$, with the metallic fraction $F_m = 1/3$ **as-**

sumed for unsorted synthesis (Section 1.5).

3.3.2 Resistance: Phonon-Scattering-Limited Model

Following Majumder et al. [12], the effective mean free path combines acoustic- and optical-phonon scattering via Matthiessen's rule:

$$\frac{1}{\lambda_{eff}} = \frac{1}{\lambda_{AC}} + \frac{1}{\lambda_{OP}}, \qquad \lambda_{AC} = \frac{400}{T/300}\ \text{nm}, \qquad \lambda_{OP} = 56\ \text{nm} \tag{3.8}$$

giving $\lambda_{eff} \approx 48.8$ nm at $T = 300$ K, using the phonon mean free path values measured by Park et al. [9].

Per-CNT resistance combines quantum, scattering, and contact terms:

$$R_{CNT} = R_Q + R_{scatter}(L) + R_{contact} \tag{3.9}$$

$$R_Q = \frac{h}{4e^2} \approx 6.45\ \text{k}\Omega, \qquad R_{scatter} = \frac{h}{4e^2} \cdot \frac{L}{\lambda_{eff}} \tag{3.10}$$

where R_Q is the fixed quantum/contact resistance per tube (four conducting channels), and $R_{contact} = 10$ **kΩ is assumed** per tube for imperfect metal-CNT contacts (Section 1.5), within the 3.2–100 kΩ range reported in the literature.

Bundle resistance per unit length is the parallel combination of N_m metallic tubes:

$$R_{bundle} = \frac{R_{CNT}}{N_m}, \qquad R_{per_length} = \frac{R_{bundle}}{L} \tag{3.11}$$

3.3.3 Inductance: Kinetic and Magnetic Components

CNT bundle inductance has two contributions: kinetic inductance, arising from electron inertia in the one-dimensional conduction channels, and magnetic inductance, of the same geometric origin as the Cu case. Kinetic inductance per tube is:

$$L_{K,tube} = \frac{h}{4e^2 v_F} \approx 8.07\ \text{nH}/\mu\text{m} \tag{3.12}$$

where $v_F = 8 \times 10^5$ m/s is the Fermi velocity in a metallic SWCNT. For N_m identical tubes in parallel, total bundle kinetic inductance per unit length is $L_{K,bundle} = L_{K,tube}/N_m$. Magnetic

inductance per unit length follows the same geometric form as the Cu case:

$$L_M = \frac{\mu_0}{2\pi} \ln\left(\frac{2H}{W} + 1\right) \tag{3.13}$$

Total bundle inductance is $L_{total} = L_{K,bundle} + L_M$. At long interconnect lengths, kinetic inductance dominates over magnetic inductance by two to three orders of magnitude, following Naeemi and Meindl [11].

3.3.4 Capacitance: Quantum and Electrostatic Components

Quantum capacitance per CNT, arising from the finite density of states at the Fermi level in a one-dimensional conductor, is:

$$C_Q = \frac{4e^2}{hv_F} \approx 193.7 \text{ aF}/\mu\text{m} \tag{3.14}$$

Total bundle quantum capacitance is $C_{Q,bundle} = C_Q \times N_m$. Electrostatic capacitance per unit length, of the same geometric origin as the Cu case, is:

$$C_E = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(2y/d)} \tag{3.15}$$

where y is the height of the bundle centre above the ground plane and d is the bundle's outer diameter. The two components combine in series per Srivastava et al. [10]:

$$\frac{1}{C_{total}} = \frac{1}{C_{Q,bundle}} + \frac{1}{C_E} \tag{3.16}$$

Since $C_{Q,bundle}$ is typically much larger than C_E at these dimensions, $C_{total} \approx C_E$ in most cases; both terms are computed explicitly at sub-14 nm nodes where C_Q becomes comparable to C_E.

3.4 Circuit-Level Simulation Setup

To validate the analytical RLC trends under real switching conditions, the extracted R, L, and C values for each node-length combination were embedded as lumped Pi-segments on the internal carry chain of two benchmark circuits: a two-stage 90 nm CMOS inverter chain and a 4-bit ripple-carry adder (RCA) built from static CMOS full adders. Both circuits use identical 90 nm

standard-cell logic and stimulus across all runs; only the interconnect Pi-segment parameters (R, L, C) are swapped between the Cu and CNT bundle cases at a given node and length, isolating the interconnect's contribution to propagation delay.

For each of the three interconnect lengths (100, 500, 1000 µm) at each of the three technology nodes (45, 14, 7 nm), stimulus periods and transient stop times were scaled proportionally to the RC time constant of that case to ensure the circuit reached settled logic levels before measurement.

3.5 Propagation Delay and Performance Metrics

Elmore delay for a driver of resistance R_{driver} feeding a distributed RC line of length L and load C_{load} is:

$$t_D = 0.69\,R_{driver}C_{load} + 0.38\,R_{line}C_{line} + 0.69\,R_{driver}C_{line} + 0.69\,R_{line}C_{load} \tag{3.17}$$

where the 0.38 coefficient applies only to the distributed self-term ($R_{line}C_{line}$); the driver-to-load and driver-to-line cross terms both carry the 0.69 coefficient, following Rabaey and Weste-Harris. A simplified form for quick comparison, using per-unit-length R and C and total length L, is:

$$t_D \approx 0.69\,R_{driver}C_{load} + 0.38\,RCL^2 \tag{3.18}$$

Power delay product (PDP) and energy delay product (EDP) are:

$$PDP = P_{dynamic} \cdot t_D = (\alpha C_{load} V_{DD}^2 f) \cdot t_D \qquad \text{[J per switching event]} \tag{3.19}$$

$$EDP = PDP \cdot t_D = P t_D^2 \qquad [\text{J} \cdot \text{s}] \tag{3.20}$$

where α is the activity factor, f the operating frequency, and V_{DD} the supply voltage. EDP is used as the principal figure of merit in Parihar and Sharma [15] and is adopted here for comparative evaluation in Chapter 4, since it captures cases where CNT bundles achieve lower energy-delay performance than Cu despite comparable or higher raw delay.

Chapter 4

Results and Discussion

4.1 Technology Node Geometry and Copper Resistivity Scaling

Table 4.1 lists the drawn and effective wire/bundle geometry at each node. Because a diffusion barrier of fixed thickness must be reserved on every sidewall of the Cu trench, the effective conducting cross-section shrinks faster than the drawn wire dimensions: at 7 nm only 5 nm of width remains available for current flow, versus 38 nm at 45 nm. The SWCNT bundle experiences an analogous effect: the number of tubes that physically fit in the same footprint, and the number of those that are metallic (roughly one in three), falls from 299 metallic tubes at 45 nm to just 24 at 7 nm.

Table 4.1: Drawn and effective wire/bundle geometry at each technology node

Node	**W (nm)**	**H (nm)**	$W_{eff,Cu}$**(nm)**	N_{total}**(CNT)**	$N_{metallic}$**(CNT)**
45 nm	45	90	38	897	299
14 nm	14	28	11	190	63
7 nm	7	14	5	72	24

Copper's size-effect-corrected resistivity (ρ_{eff}), combining Fuchs-Sondheimer surface scattering with Mayadas-Shatzkes grain-boundary scattering, degrades sharply and non-linearly as the node shrinks (Table 4.2). Relative to bulk copper (1.68 $\mu\Omega$cm), ρ_{eff} is only 1.45$\times$ higher at 45 nm but reaches 10.02$\times$ bulk at 7 nm.

Table 4.2: Copper effective resistivity scaling due to combined surface and grain-boundary scattering

Node	ρ_{FS} Mult.	ρ_{MS} Mult.	ρ_{eff} (μΩcm)	vs. Bulk Cu
45 nm	1.163	1.247	2.43	1.45×
14 nm	2.601	1.769	7.73	4.60×
7 nm	3.984	2.514	16.83	10.02×

4.2 Resistance, Inductance, and Capacitance: Cu vs. SWCNT Bundles

4.2.1 Resistance

Table 4.3 summarises extracted resistance across all node-length combinations. At every node the CNT/Cu resistance ratio is essentially independent of wire length, since bundle resistance is dominated by the fixed quantum resistance ($R_Q = h/4e^2 \approx 6.45$ kΩ per tube) rather than the length-dependent scattering term. The ratio itself falls steadily with node: from 50.1× at 45 nm to 7.4–7.5× at 14 nm, and just 2.1× at 7 nm.

Table 4.3: Total interconnect resistance, copper vs. SWCNT bundle, across node and length

Node	Length (μm)	Cu R (Ω)	CNT R (Ω)	CNT/Cu Ratio
45 nm	100	877.7	43,991	50.1×
45 nm	500	4,389	219,735	50.1×
45 nm	1000	8,777	439,415	50.1×
14 nm	100	28,006	208,783	7.5×
14 nm	500	140,028	1,042,870	7.4×
14 nm	1000	280,057	2,085,479	7.5×
7 nm	100	258,858	548,055	2.1×
7 nm	500	1,294,291	2,737,535	2.1×
7 nm	1000	2,588,583	5,474,384	2.1×

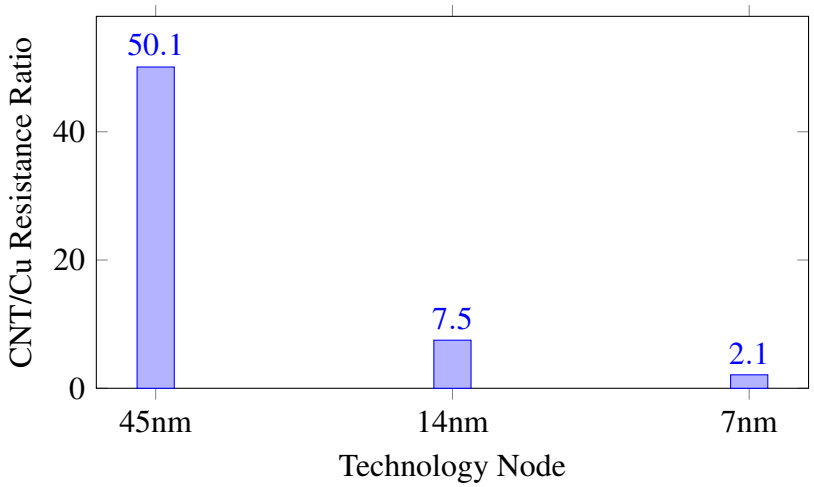

Figure 4.1: CNT/Cu resistance ratio versus technology node, showing the gap closing sharply as Cu's own resistivity degrades

4.2.2 Inductance

CNT inductance is dominated by kinetic inductance rather than the conventional magnetic inductance that governs copper (Table 4.4). Because kinetic inductance scales inversely with the number of metallic tubes, the CNT/Cu ratio grows sharply as the node shrinks and $N_{metallic}$ falls, from roughly 13–17× at 45 nm to 141–174× at 7 nm. This trend runs opposite to the resistance trend: as CNTs become more resistance-competitive at smaller nodes, their inductance disadvantage simultaneously worsens.

Table 4.4: CNT/Cu inductance ratio by node and length (compressed from full RLC data)

Node	**100** µm	**500** µm	**1000** µm
45 nm	17.2×	14.3×	13.4×
14 nm	71.3×	60.5×	56.8×
7 nm	174.3×	149.3×	140.6×

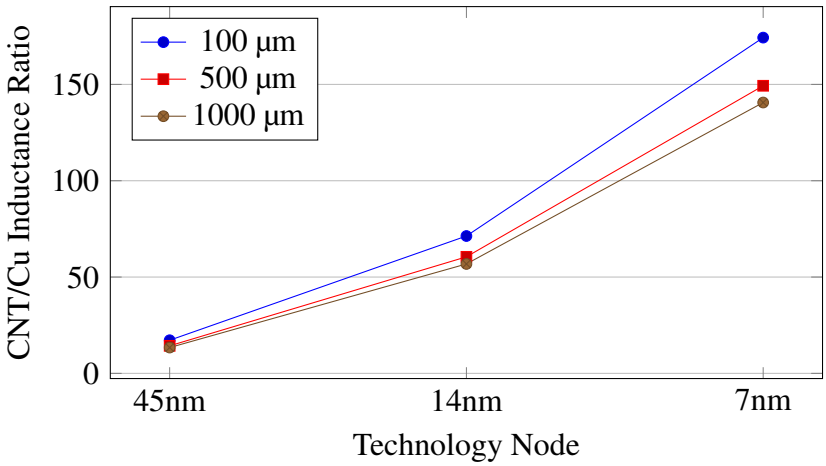

Figure 4.2: CNT/Cu inductance ratio versus node, at each interconnect length. Kinetic inductance grows inversely with $N_{metallic}$

4.2.3 Capacitance

Capacitance shows the most consistent trend of the three parameters (Table 4.5): the CNT bundle is 5.3–5.6× more capacitive than copper at every node and length studied, attributable to quantum capacitance, an intrinsic per-channel property of one-dimensional conductors that does not shrink away with geometric scaling.

Table 4.5: Total interconnect capacitance, copper vs. SWCNT bundle, across node and length

Node	**Length** (µm)	**Cu C (fF)**	**CNT C (fF)**	**CNT/Cu Ratio**
45 nm	100	3.28	17.36	5.3×
45 nm	500	16.41	86.80	5.3×
45 nm	1000	32.83	173.61	5.3×
14 nm	100	2.46	13.56	5.5×
14 nm	500	12.32	67.81	5.5×
14 nm	1000	24.64	135.62	5.5×
7 nm	100	2.16	11.99	5.6×
7 nm	500	10.79	59.95	5.6×
7 nm	1000	21.59	119.89	5.6×

4.3 RC Time Constant Analysis

The RC product ($\tau = RC$) captures the combined effect of the resistance and capacitance trends above, and is the dominant driver of the propagation delays observed in Section 4.5. The CNT/Cu τ ratio shrinks dramatically, from about 265× at 45 nm to roughly 41× at 14 nm and 11.7–11.8× at 7 nm, almost entirely because copper's resistance degrades so rapidly at advanced nodes, not because the CNT bundle itself improves.

Table 4.6: RC time constant, copper vs. SWCNT bundle, at 45 nm, 14 nm, and 7 nm

Node	**Length** (μm)	**Cu τ (ps)**	**CNT τ (ps)**	**CNT/Cu Ratio**
45 nm	100	2.88	763.6	265×
45 nm	500	72.0	19,074	265×
45 nm	1000	288	76,297	265×
14 nm	100	68.9	2,832	41.1×
14 nm	500	1,725	70,726	41.0×
14 nm	1000	6,900	282,904	41.0×
7 nm	100	559	6,571	11.8×
7 nm	500	13,972	164,145	11.7×
7 nm	1000	55,887	656,291	11.7×

4.4 Circuit-Level Simulation Results

Extracted R, L, and C values were embedded as LTspice parameters and exercised in two test circuits at fixed 90 nm CMOS device sizing, so any change in delay or signal quality is attributable to the interconnect alone: a two-stage inverter chain, isolating the wire's own contribution to propagation delay, and a 4-bit ripple-carry adder (RCA), in which the interconnect is inserted into the internal carry chain so delay accumulates across three cascaded segments. The complete set of nine simulated node-length combinations is presented below, moving from the mildest case (45 nm, 100 μm) to the most demanding (7 nm, 1000 μm), so that the progressive divergence between Cu and CNT waveforms is visible directly, not only in the summary delay tables.

4.4.1 CMOS Inverter Chain: Propagation Delay

Table 4.7 reports the approximate 50%-crossing propagation delay for each case. The worst relative case occurs at 45 nm/1000 µm, where CNT delay (≈450 ns) is roughly 120× slower than copper (≈3.7 ns). At the opposite extreme, at 7 nm/1000 µm both materials fall into the microsecond range (≈3.48 µs for Cu, ≈10.87 µs for CNT), indicating a fundamental interconnect bottleneck that neither material resolves.

Table 4.7: Two-stage CMOS inverter-chain propagation delay, 90 nm devices with scaled interconnect

Node	**Length** (µm)	**Cu Delay (≈50%)**	**CNT Delay (≈50%)**
45 nm	100	∼5.5 ns	∼11.5 ns
45 nm	500	∼4.7 ns	∼169 ns
45 nm	1000	∼3.7 ns	∼450 ns
14 nm	100	∼5.8 ns	∼59 ns
14 nm	500	∼33 ns	∼0.75 µs
14 nm	1000	∼114 ns	∼6.3 µs
7 nm	100	∼13.8 ns	∼89 ns
7 nm	500	∼960 ns	∼3.5 µs
7 nm	1000	∼3.48 µs	∼10.87 µs

45 nm node: baseline behaviour

Figure 4.3 shows the full simulation hierarchy followed at every node: for each of the three lengths studied (100 µm, 500 µm, 1000 µm), four waveforms were extracted – Cu inverter, CNT inverter, Cu RCA, and CNT RCA – giving nine length/node cells and thirty-six waveform captures across the full study.

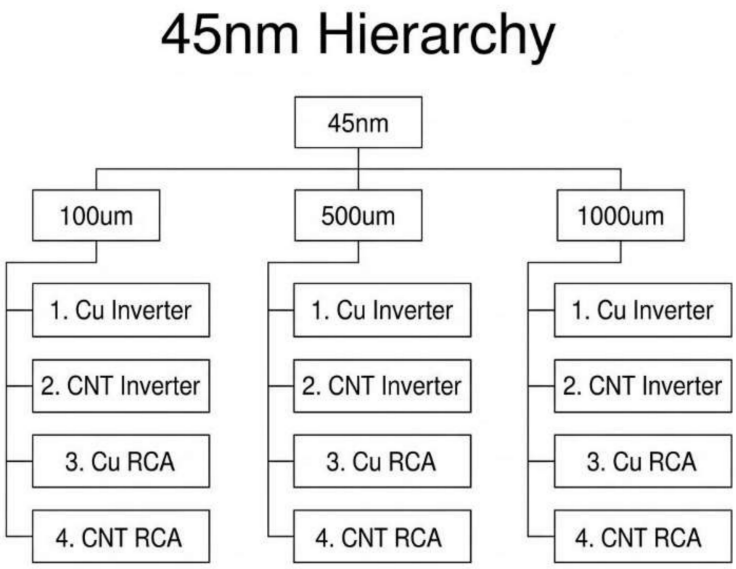

Figure 4.3: Simulation hierarchy at the 45 nm node: three interconnect lengths, each exercised through Cu inverter, CNT inverter, Cu RCA, and CNT RCA test circuits

At 45 nm and 100 µm, both interconnects still produce clean, near-identical square-edged transitions for the inverter chain and clean pulse trains for the RCA carry chain; Cu and CNT are barely distinguishable at this shortest, most relaxed case.

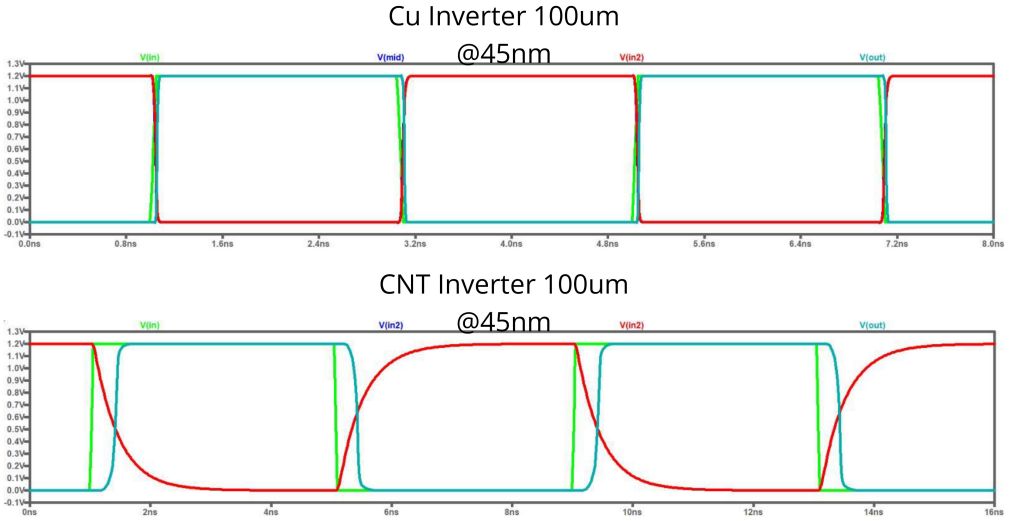

Figure 4.4: Two-stage inverter-chain transient response at 45 nm, 100 µm: copper (top) and SWCNT bundle (bottom), both producing clean, near-identical square-edged transitions at this baseline case

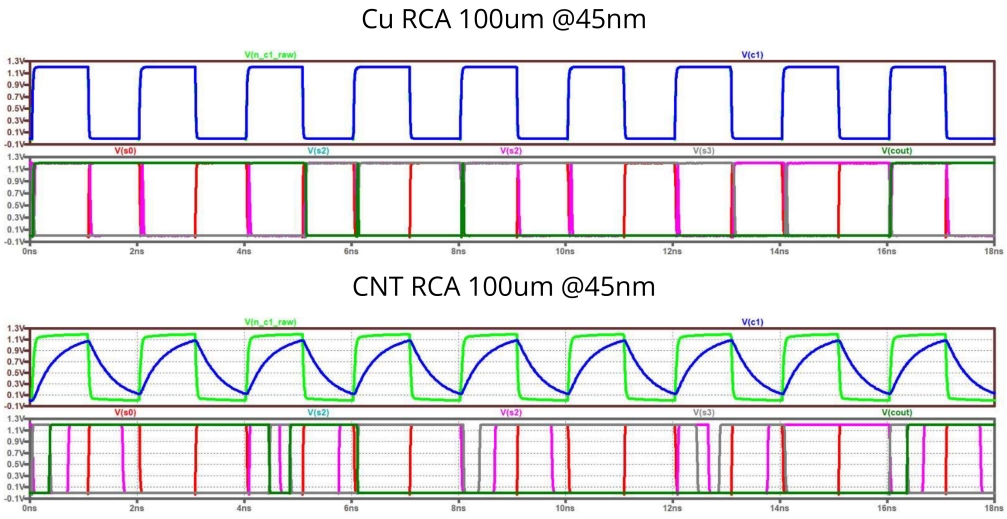

Figure 4.5: 4-bit RCA carry-chain waveforms at 45 nm, 100 µm: copper (top) and SWCNT bundle (bottom), both showing clean digital transitions at this baseline node/length combination

As length increases to 500 µm at the same node, the Cu inverter response stays essentially unchanged, while the CNT inverter output visibly slows into a rounded exponential-like transition, and the CNT RCA carry chain begins to show clear slowing between pulses.

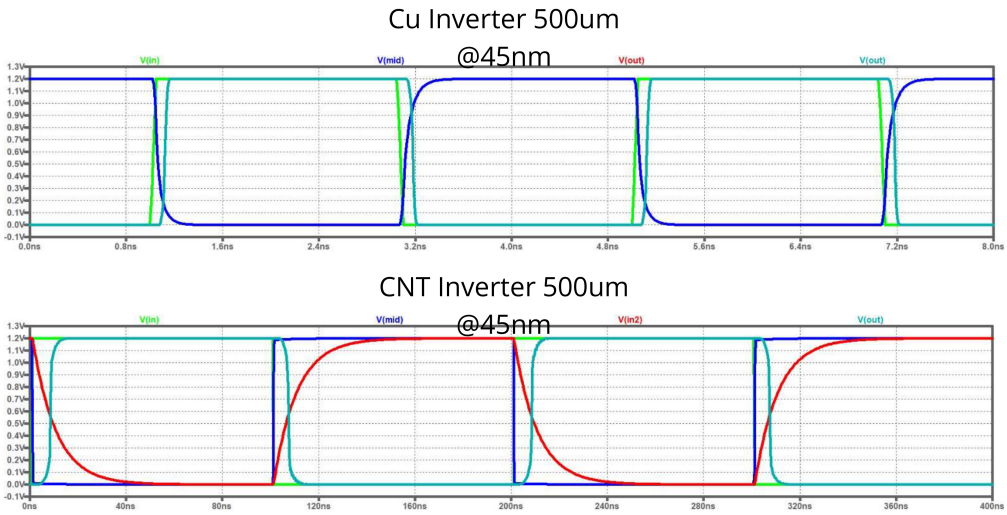

Figure 4.6: Two-stage inverter-chain transient response at 45 nm, 500 µm: copper (top) stays sharp while the SWCNT bundle (bottom) rounds into an exponential-like transition

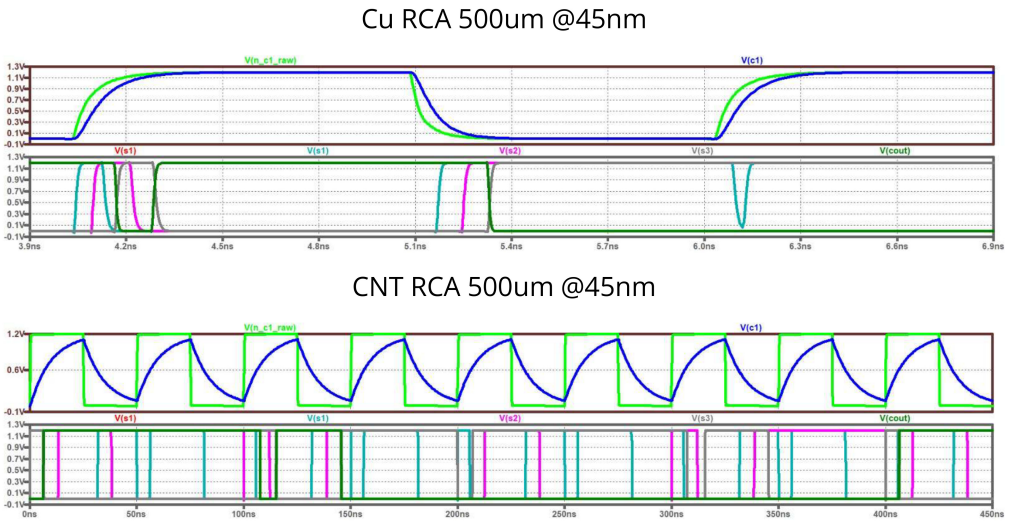

Figure 4.7: 4-bit RCA carry-chain waveforms at 45 nm, 500 µm: copper (top) stays clean while the CNT bundle (bottom) shows visible slowing between pulses

By 1000 µm at 45 nm, the divergence is most pronounced: the CNT inverter transition stretches out substantially further, and the CNT RCA carry chain shows the largest slowing observed at this node.

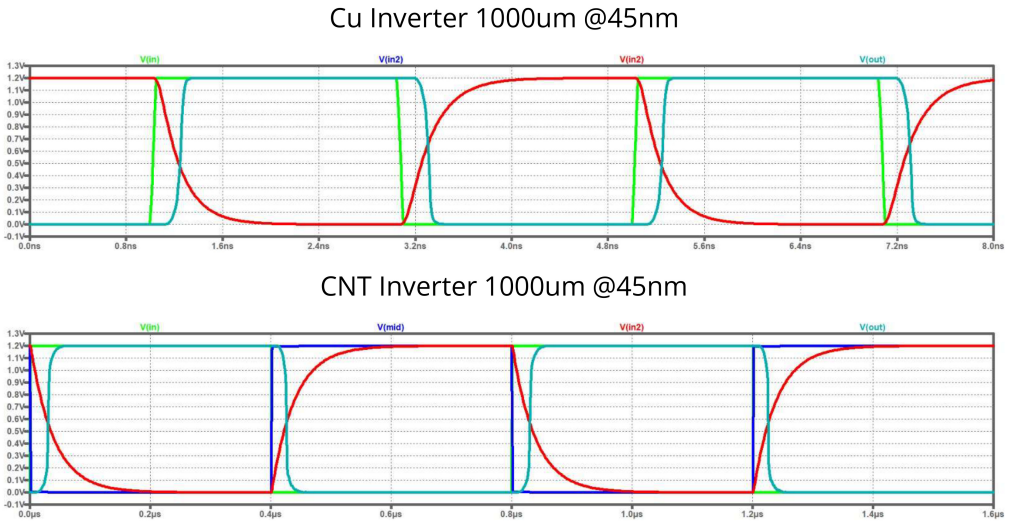

Figure 4.8: Two-stage inverter-chain transient response at 45 nm, 1000 µm: copper (top) versus SWCNT bundle (bottom), the largest node-relative delay gap observed in the study ($\approx 120\times$)

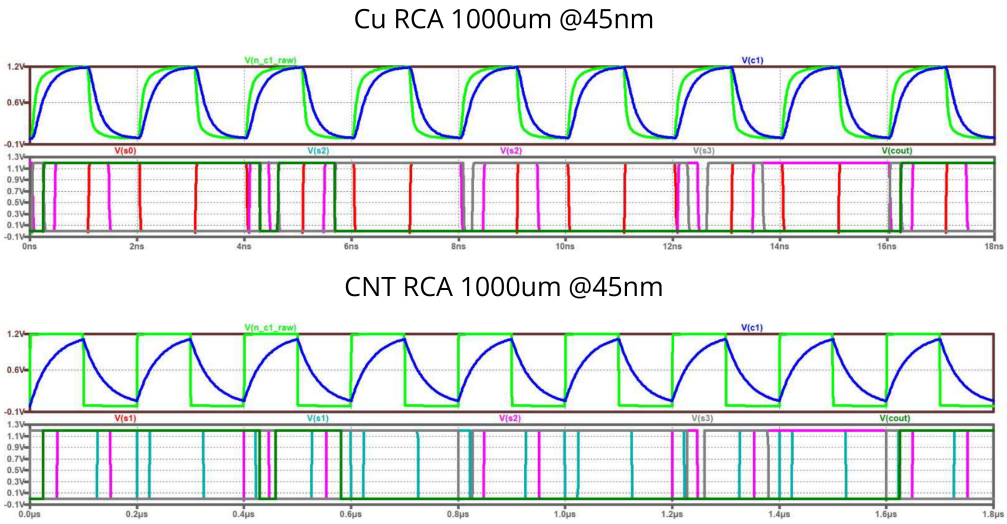

Figure 4.9: 4-bit RCA carry-chain waveforms at 45 nm, 1000 µm: copper (top) remains clean while the CNT bundle (bottom) shows pronounced inter-pulse slowing

14 nm node: divergence begins

At 14 nm the two materials start to separate even at the shortest length. Table 4.7 shows CNT inverter delay already an order of magnitude above Cu at 100 µm (∼59 ns vs. ∼5.8 ns), and by 1000 µm the CNT delay has climbed into the microsecond range while Cu remains in the sub-microsecond range.

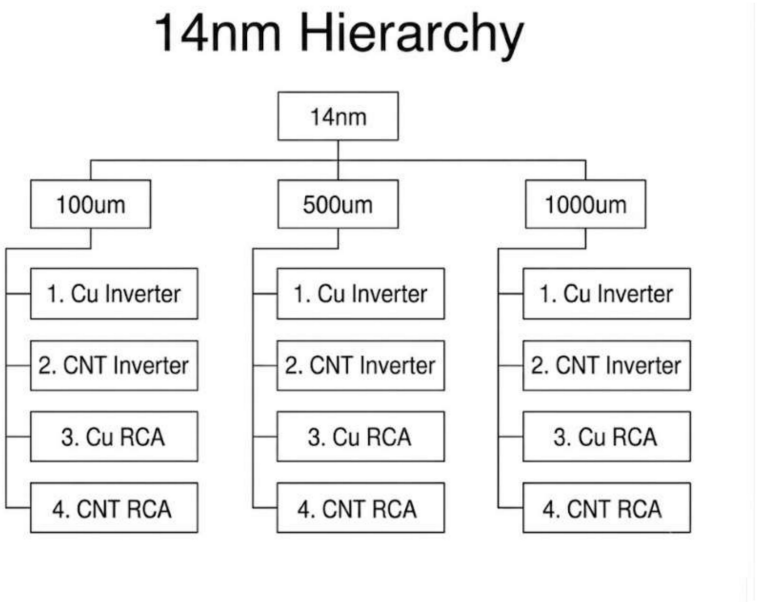

Figure 4.10: Simulation hierarchy at the 14 nm node, following the same length/circuit structure as Figure 4.3

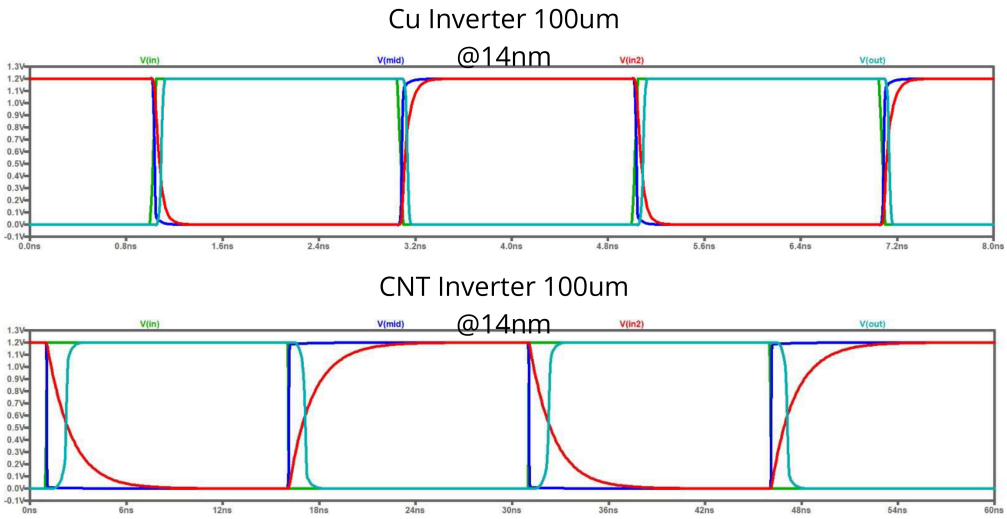

Figure 4.11: Two-stage inverter-chain transient response at 14 nm, 100 µm: the CNT inverter transition (bottom) is already visibly rounded compared to Cu's sharp edge (top)

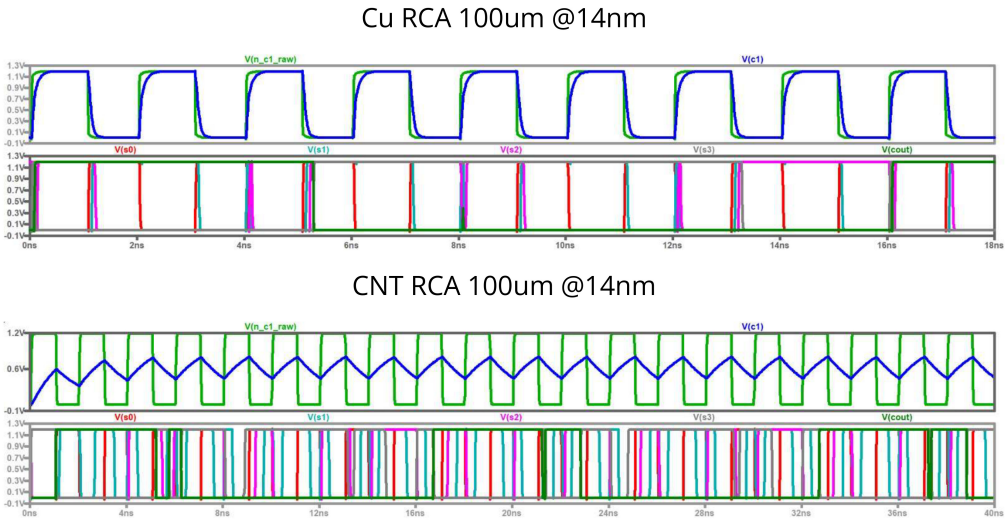

Figure 4.12: 4-bit RCA carry-chain waveforms at 14 nm, 100 µm: copper (top) and SWCNT bundle (bottom); the carry chain remains functional for both materials at this length

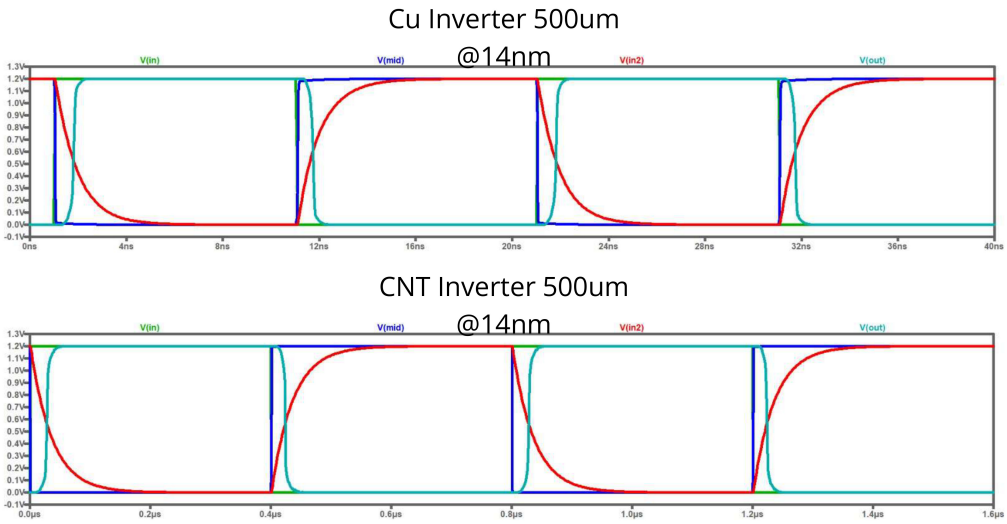

Figure 4.13: Two-stage inverter-chain transient response at 14 nm, 500 µm: copper (top) versus SWCNT bundle (bottom), delay climbing toward the sub-microsecond range for CNT

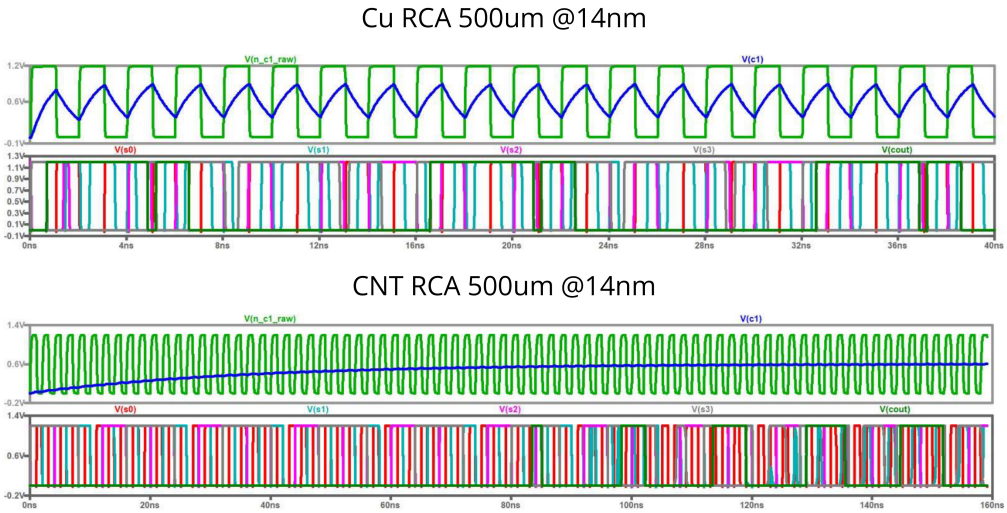

Figure 4.14: 4-bit RCA carry-chain waveforms at 14 nm, 500 µm: CNT RCA carry pulses (bottom) become visibly ragged and unevenly spaced relative to copper's regular pulse train (top)

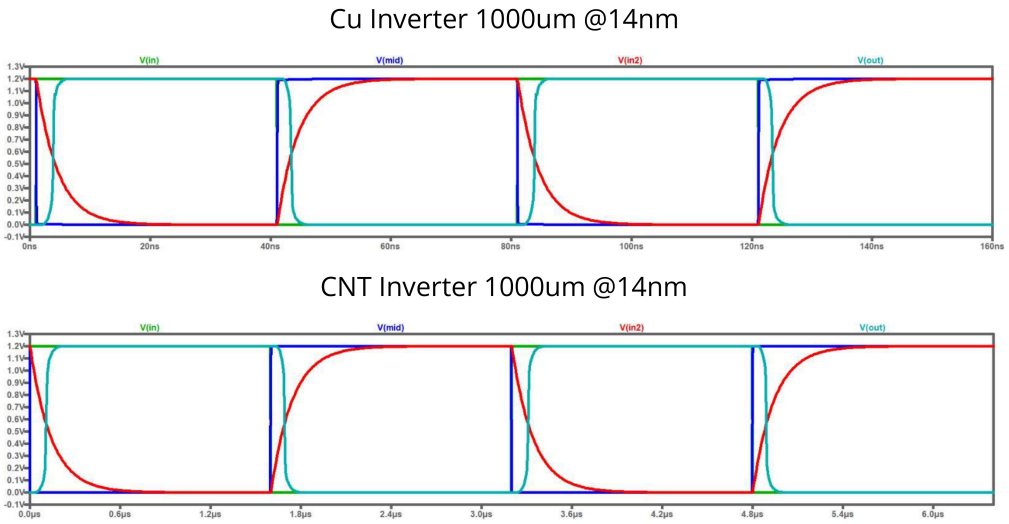

Figure 4.15: Two-stage inverter-chain transient response at 14 nm, 1000 µm: copper (top) remains sub-microsecond while the CNT bundle (bottom) reaches into the microsecond range

At 14 nm/1000 µm (bundle resistance ≈2.09 MΩ), the RCA internal carry waveform becomes essentially a sawtooth rather than a square pulse, a signal-integrity failure mode not observed for copper at any node or length studied.

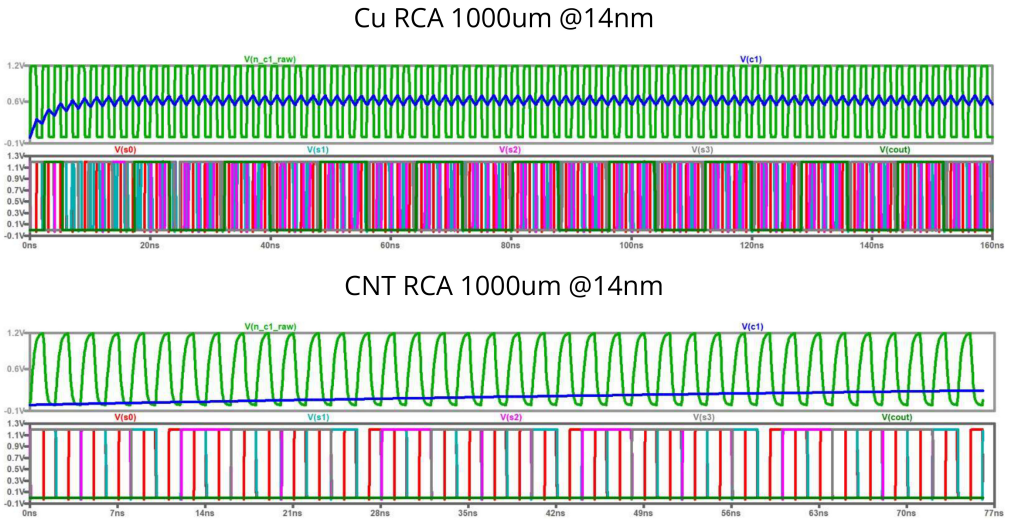

Figure 4.16: 4-bit RCA carry-chain waveform at 14 nm, 1000 µm, SWCNT bundle interconnect: the internal carry node degrades into a sawtooth, non-digital signal, functionally broken for reliable logic operation

7 nm node: both materials under stress

At 7 nm, copper itself begins to show measurable slowing at longer lengths (Cu delay rises from ∼13.8 ns at 100 µm to ∼3.48 µs at 1000 µm), confirming that the interconnect bottleneck

is no longer CNT-specific at this node. The CNT bundle remains slower throughout, but the relative gap has narrowed to the smallest margin observed in the study (Section 4.2.1).

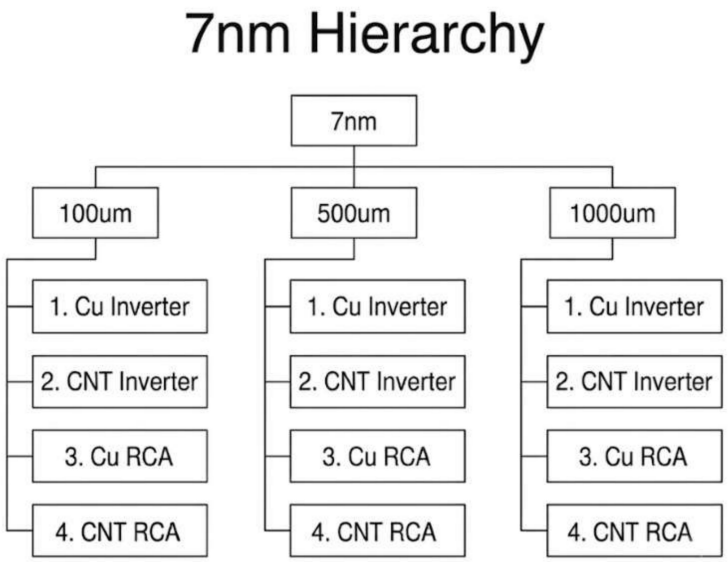

Figure 4.17: Simulation hierarchy at the 7 nm node, following the same length/circuit structure as Figure 4.3

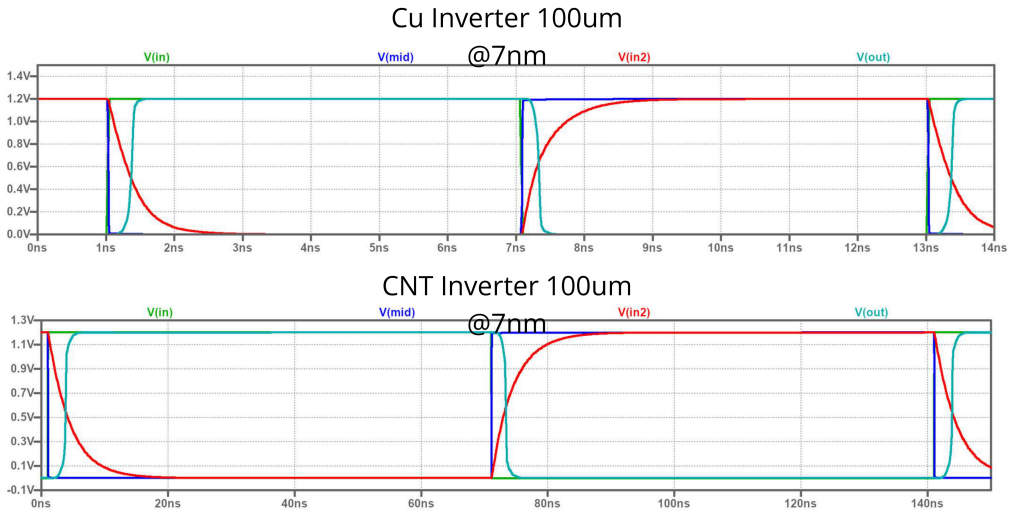

Figure 4.18: Two-stage inverter-chain transient response at 7 nm, 100 µm: Cu (top) shows early signs of edge rounding at this node, while CNT (bottom) is further degraded but the two are visibly closer than at larger nodes

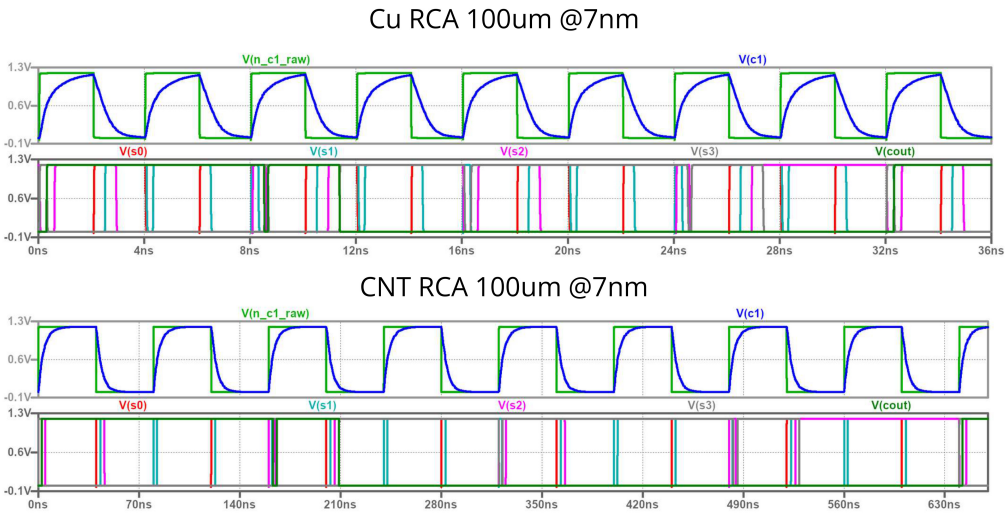

Figure 4.19: 4-bit RCA carry-chain waveforms at 7 nm, 100 µm: copper (top) and SWCNT bundle (bottom), both functional with Cu now also showing visible rounding

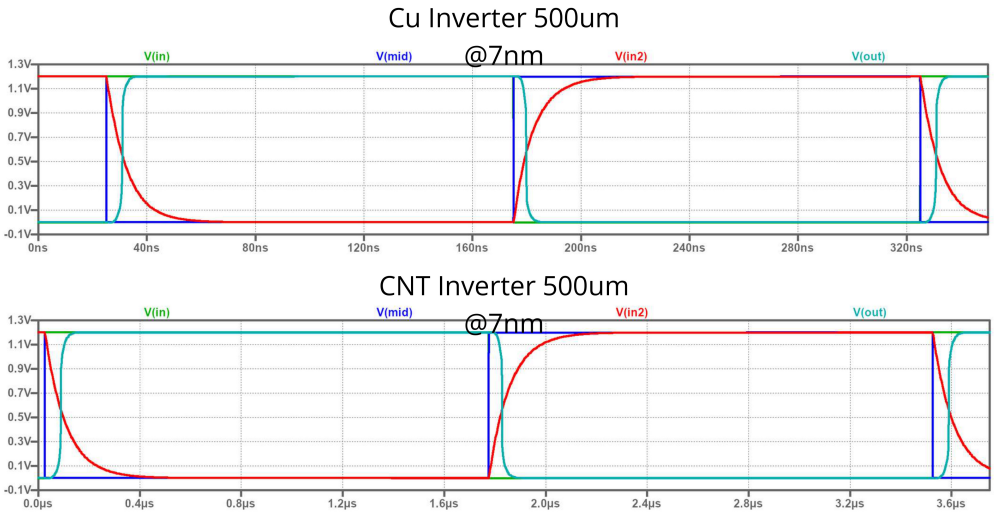

Figure 4.20: Two-stage inverter-chain transient response at 7 nm, 500 µm: both Cu (top) and CNT (bottom) show pronounced slewing at this length

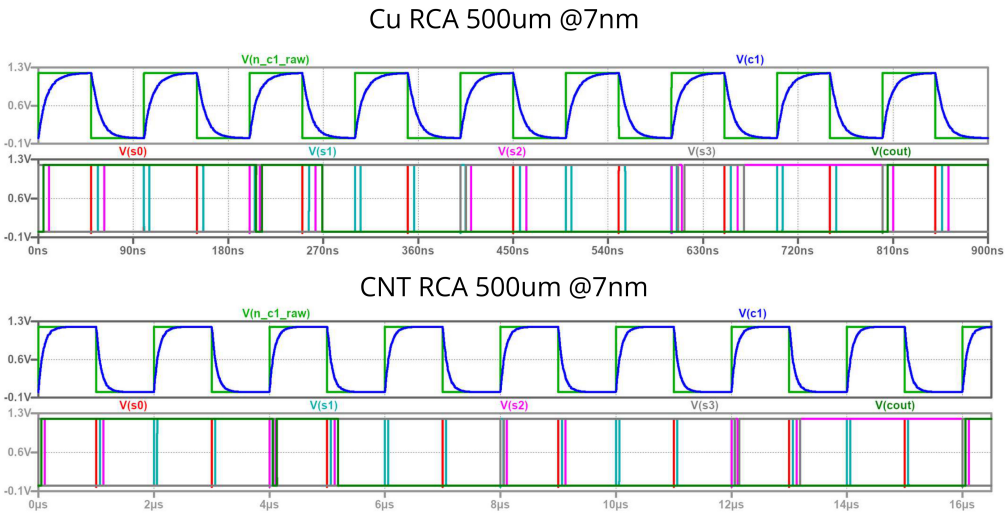

Figure 4.21: 4-bit RCA carry-chain waveforms at 7 nm, 500 µm: both materials show pronounced slewing, with CNT (bottom) approximately 12× slower than Cu (top) on the RCA carry chain (Table 4.8)

The worst-case combination for both materials occurs at 7 nm/1000 µm, where both Cu (≈3.48 µs) and CNT (≈10.87 µs) fall into the microsecond range, indicating a fundamental interconnect bottleneck that neither material resolves.

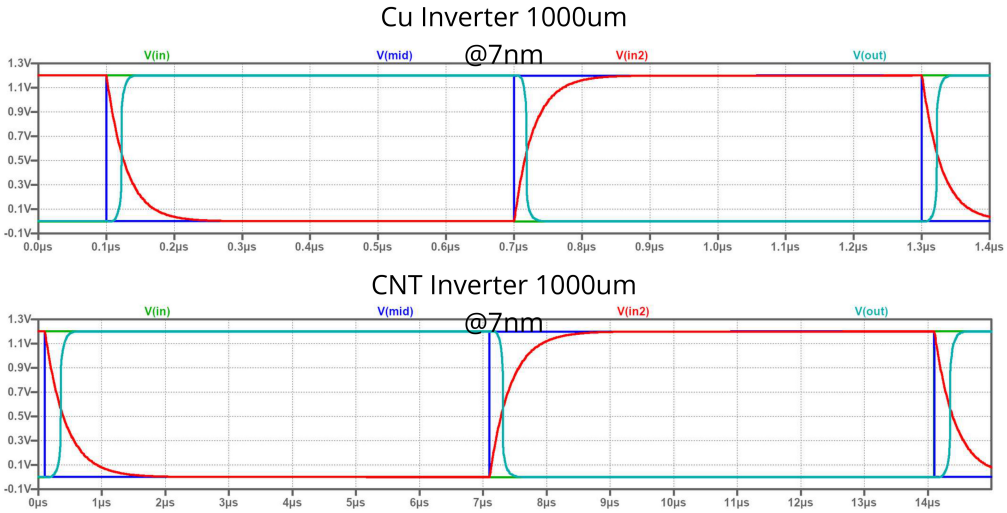

Figure 4.22: Two-stage inverter-chain transient response at 7 nm, 1000 µm: copper (top, settling at ≈3.5 µs) versus SWCNT bundle (bottom, settling at ≈11 µs), the worst-case bottleneck for both materials

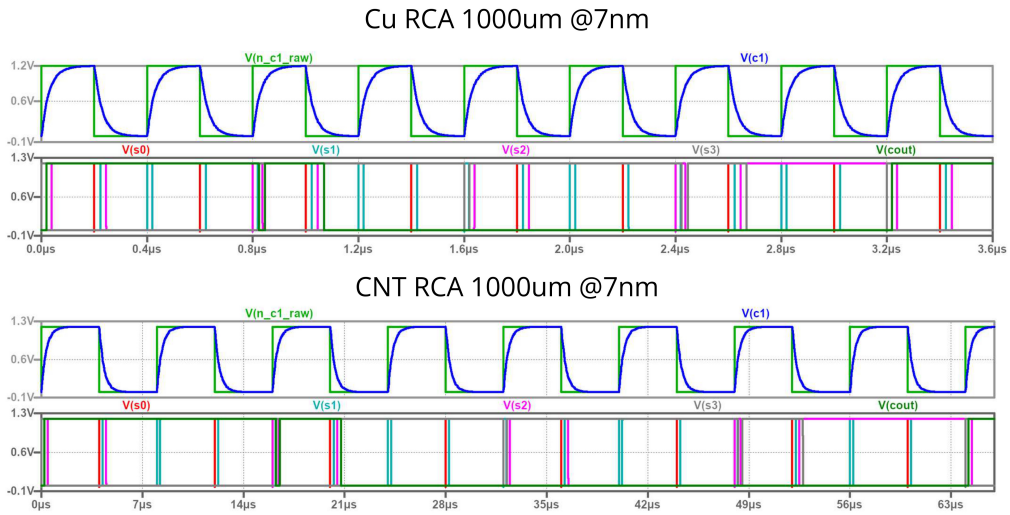

Figure 4.23: 4-bit RCA carry-chain waveforms at 7 nm, 1000 µm, the most demanding node-length combination studied: copper (top) and SWCNT bundle (bottom) both push into the microsecond delay regime

4.4.2 4-Bit Ripple-Carry Adder: Signal Integrity

The RCA results (Table 4.8) show that the CNT bundle's resistance and capacitance penalties translate directly into degraded logic-level signal quality, not merely added delay.

Table 4.8: Ripple-carry-adder carry-chain propagation time and signal quality, 90 nm logic with scaled interconnect

Node	**Length (µm)**	**Cu Carry Time**	**CNT Carry Time**	**Signal Quality**
45 nm	100	∼11.8 ns	∼10.9 ns	Both clean
45 nm	500	∼6.5 ns/bit	∼89 ns/bit	CNT visibly slowed
45 nm	1000	∼6.5 ns	∼286 µs	CNT severely degraded
14 nm	100	∼18 ns	∼660 ns	CNT over-damped
14 nm	500	∼33 ns	∼53.7 ns	CNT distorted
14 nm	1000	∼26 ns	∼66 ns	CNT sawtooth / broken
7 nm	100	∼24.6 ns	∼660 ns	Cu degraded, functional
7 nm	500	∼708 ns	∼8.4 µs	CNT ∼12× slower
7 nm	1000	∼3.48 µs	∼60.5 µs	Both marginal

4.5 Scaling Trends: Length and Node Dependence

At a fixed node, both materials' resistance scales essentially linearly with interconnect length (Table 4.9): copper because ρ is length-independent, and the CNT bundle because the phonon-scattering term already dominates over the fixed quantum/contact resistance once length exceeds $\lambda_{eff} \approx 49$ nm.

Table 4.9: Resistance vs. interconnect length at the 7 nm node

Length (μm)	**Cu R (Ω)**	**Growth**	**CNT R (Ω)**	**Growth**
100	258,858	1×	548,055	1×
500	1,294,291	5.0×	2,737,535	5.0×
1000	2,588,583	10.0×	5,474,384	9.99×

At a fixed length, the two materials scale very differently with node (Table 4.10). From 45 nm to 7 nm, Cu resistance grows 295× at 1000 μm, driven by the resistivity rise in Section 4.1. Over the same range, CNT bundle resistance grows only 12.5×, governed by the shrinking bundle population ($N_{metallic}$ falling from 299 to 24) rather than any node-dependent change in scattering physics.

Table 4.10: Resistance vs. technology node at a fixed interconnect length of 1000 μm

Node	**Cu R (kΩ)**	**Growth**	**CNT R (kΩ)**	**Growth**
45 nm	8.78	1×	439.4	1×
14 nm	280.1	31.9×	2,085	4.7×
7 nm	2,589	295×	5,474	12.5×

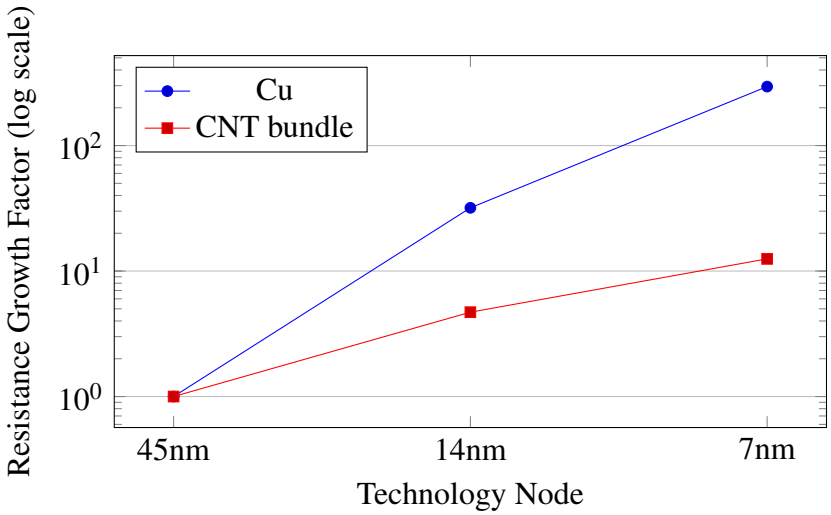

Figure 4.24: Resistance growth factor (normalized to 45 nm) versus technology node at 1000 µm, log scale. Cu's scattering-driven degradation (295×) far outpaces CNT's packing-limited growth (12.5×)

4.6 Crossover Analysis

Table 4.11 identifies the approximate interconnect length at which Cu and CNT-bundle resistance become equal at each node. At 45 nm the crossover length is far beyond any practical on-chip interconnect ($\gg$10 mm). The crossover length then shrinks steadily with node, reaching roughly 2.5 mm at 14 nm and only about 400 µm at 7 nm, squarely within the range of practical intermediate/global interconnect lengths.

Table 4.11: Approximate interconnect length at which copper and CNT-bundle resistance converge, by node

Node	**Length for** $R_{Cu} \approx R_{CNT}$	**Status**
45 nm	$\gg$10 mm	Cu always better
14 nm	$\sim$2.5 mm	Cu better, gap narrowing
7 nm	$\sim$400 µm	Near-competitive at 500 µm

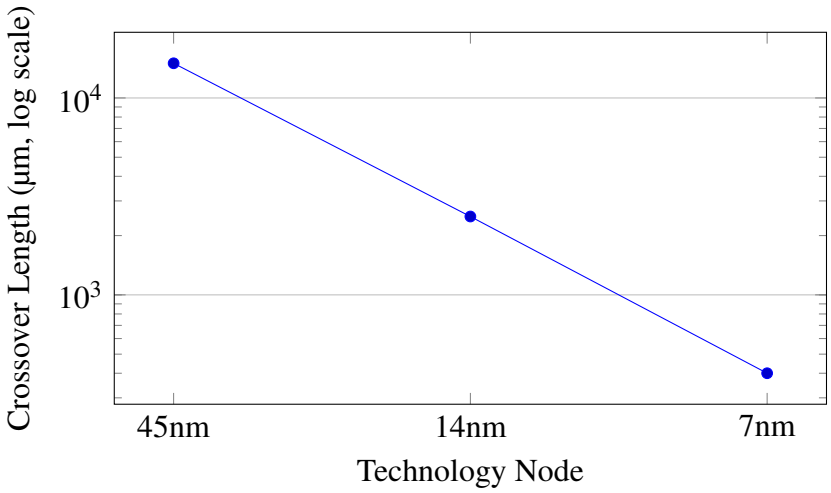

Figure 4.25: Approximate Cu-CNT resistance crossover length versus technology node, log scale. The crossover length collapses from far beyond practical wiring at 45 nm to within global-tier range (∼400 µm) at 7 nm

4.7 Discussion

4.7.1 Physical Origins of Copper's Degradation at Advanced Nodes

Copper's resistivity blow-up follows directly from the two scattering mechanisms compounding as the effective wire width shrinks toward and below copper's electron mean free path ($\lambda_{Cu} \approx 40$ nm). At 45 nm, the effective width (38 nm) is still comparable to λ_{Cu}, surface scattering is mild, and the grain-boundary parameter α (≈0.17) sits in a sub-linear regime, so ρ_{eff} is only 45% above bulk. By 14 nm, the effective width (11 nm) is well below λ_{Cu}, surface scattering becomes strong, and α (≈0.55) moves deep into the grain-boundary-saturation regime. At 7 nm the effective width (5 nm) is comparable to a single scattering mean free path, surface specularity is effectively zero, and the two effects compound rather than simply add, producing the observed 10×-bulk resistivity.

4.7.2 Physical Origins of CNT Bundle Behaviour

The CNT bundle's total resistance can be written as $R_{total} = (R_Q + R_Q \cdot (L/\lambda_{eff}) + R_{contact})/N_{metallic}$. At interconnect lengths of 100 µm and above, the scattering term dominates the numerator and is essentially fixed by phonon physics, it does not change with technology node. What changes with node is $N_{metallic}$, falling from 299 to 24 across the nodes studied, so the bundle's resistance scales only through this geometric packing factor. This is the mechanistic reason CNT

resistance scales far more mildly with node (12.5×) than copper's scattering-driven resistivity (295×): the CNT penalty is architectural rather than a fundamental materials degradation.

4.7.3 Trade-offs and Engineering Implications

The results point to a technology-node-dependent crossover rather than a universal winner. At 45 nm, copper's resistance advantage (50×) is overwhelming. At 14 nm the advantage has fallen to 7–8×. At 7 nm, copper's resistance advantage has collapsed to only 2.1×, and for interconnects longer than roughly 400–500 μm the two materials are approximately resistance-matched, even though copper still wins decisively on inductance (140–174× lower at 7 nm) and capacitance (5.6× lower).

CNT bundles' principal liabilities at advanced nodes are reactive rather than resistive: kinetic inductance and quantum capacitance are both intrinsic, quantum-mechanical properties of one-dimensional conduction and cannot be engineered away the way copper's electromigration or barrier-thickness problems can. They can only be diluted by increasing the metallic fraction F_m, itself a difficult fabrication and sorting problem. The RCA results reinforce that these penalties are not merely a delay inconvenience: at several node/length combinations the CNT interconnect's RC loading was severe enough to distort the carry-chain waveform into a non-digital, sawtooth-like signal.

For near-term nodes (7 nm and below), copper, typically paired with cobalt or ruthenium barrier/liner metals, remains the more practical choice for local and intermediate interconnects. CNT bundles are most credible as a medium-term candidate for global interconnects longer than about 500 μm at 7 nm and below, contingent on reducing the metal-nanotube contact resistance and improving the sorted metallic fraction toward $F_m \to 1$. Even so, because RC delay depends on both terms, matching or beating copper's resistance alone is not sufficient: the 5.3–5.6× capacitance penalty means resistance and capacitance must be improved together for CNT interconnects to offer a genuine delay advantage.

4.8 Evaluation Summary

Table 4.12: Overall evaluation summary of the Cu vs. CNT bundle comparison

Component	**Metric**	**Values**	**Interpretation**	**Engineering Significance**
Resistance	CNT/Cu ratio	$50.1\times \rightarrow 2.1\times$ (45 nm $\rightarrow$ 7 nm)	Gap narrows sharply as Cu resistivity degrades	CNT becomes resistance-competitive only at advanced nodes
Inductance	CNT/Cu ratio	$13\times \rightarrow 174\times$ (45 nm $\rightarrow$ 7 nm)	Kinetic inductance grows as $N_{metallic}$ falls	CNT's reactive penalty worsens exactly where its resistance improves
Capacitance	CNT/Cu ratio	5.3–5.6×, all cases	Fixed quantum capacitance overhead	Consistent penalty regardless of node or length
Crossover length	$R_{Cu} \approx R_{CNT}$	$\gg$10 mm (45 nm) $\rightarrow \sim$400 µm (7 nm)	Practical crossover only emerges at 7 nm	Identifies the node/length window where CNT becomes viable
Circuit delay	Inverter chain, worst case	3.48 µs (Cu) vs. 10.87 µs (CNT) at 7 nm/1000 µm	Both materials bottleneck at this combination	Neither material alone resolves long, advanced-node interconnects
Signal integrity	RCA carry chain	Sawtooth waveform, 14 nm/1000 µm CNT	CNT RC loading distorts logic levels	Confirms delay penalty translates into functional failure, not just slowdown

The comparison confirms that copper remains the superior interconnect material at relaxed nodes, while SWCNT bundles narrow the gap substantially, though not completely, at 7 nm, positioning them as a promising but contact-resistance-limited alternative for future global-tier VLSI interconnects.

Chapter 5

Conclusion and Future Scope

5.1 Introduction

This chapter pulls together the main findings of the comparative Cu-CNT interconnect study, looks at where the current analytical and simulation framework falls short, and points toward where the work could go next.

5.2 Conclusion

5.2.1 The Core Trade-off

The study points to a clear crossover between Cu and SWCNT bundle interconnects, one that shifts depending on the technology node:

- **At 45 nm:** Cu wins outright, with resistance roughly 50× lower than the CNT bundle, and it stays the better choice across every interconnect length studied.
- **At 14 nm:** Cu is still ahead, but the margin drops to roughly 7.5× – a sign that Cu's advantage is eroding fast.
- **At 7 nm:** Cu only just edges out CNT, at about 2.1× lower resistance, and by this point both materials are struggling at millimetre-scale lengths.

Copper's strengths lie in its low resistance and negligible inductance at relaxed nodes (45 nm), its minimal capacitance, and a manufacturing process that's well proven at every node studied here. Its weak point is the sharp, non-linear rise in resistivity below 14 nm – driven

by surface and grain-boundary scattering along with barrier-layer erosion – which together produce a 295× jump in resistance between 45 nm and 7 nm at 1 mm length.

CNT bundles tell the opposite story: only a 12.5× increase in resistance across that same 45 nm to 7 nm range, better scaling with node, higher current-carrying capacity per tube, and no electromigration risk thanks to covalent C-C bonding. Their main drawback is the fixed quantum and contact resistance overhead, which makes short CNT bundles quite resistive – roughly 50× higher than Cu at 100 µm/45 nm – on top of noticeably higher kinetic inductance and capacitance that runs 5.3–5.6× higher than Cu's across the board, a consequence of the unavoidable quantum capacitance term.

5.2.2 The Contact Resistance Bottleneck

The biggest limitation facing CNT bundle interconnects is the quantum/contact resistance, $R_Q \approx 6.45$ kΩ per tube – a fixed overhead that can't be engineered away without changing the underlying quantum transport mechanism itself. This overhead dominates at short lengths but matters proportionally less as the interconnect gets longer, where scattering resistance takes over instead. In practice, the only real lever available is better metal-CNT contact fabrication, which lowers the $R_{contact}$ term assumed throughout this work.

5.3 Research Contributions

This project makes the following contributions:

- A self-consistent RLC modelling framework applied consistently across three technology nodes (45 nm, 14 nm, 7 nm) and three interconnect lengths (100, 500, 1000 µm), pairing the latest validated Cu resistivity model with the updated phonon-scattering CNT bundle model.
- Clear, explicit treatment of the two most sensitive modelling assumptions – the metallic tube fraction $F_m = 1/3$ and the metal-CNT contact resistance $R_{contact} = 10\,\text{k}\Omega$ – rather than folding them in as fixed constants.
- Circuit-level validation of the analytical RLC trends via transient simulation of a CMOS inverter chain and a 4-bit ripple-carry adder, showing that CNT-loaded circuits at advanced nodes and longer lengths produce distorted, non-digital waveforms consistent with the elevated R and L values predicted analytically.

- A quantified crossover analysis pinning down exactly which node-length combinations bring CNT bundle resistance close to, but not below, Cu resistance.

5.4 Limitations of the Current Study

- **Fixed Assumption Values:** The metallic tube fraction ($F_m = 1/3$) and contact resistance ($R_{contact} = 10$ kΩ) are treated as single-point assumptions rather than measured or swept values, even though the literature shows both vary considerably depending on synthesis and contact fabrication method.
- **Isolated Single-Line Analysis:** All calculations apply to a single, isolated signal line; coupling to adjacent lines isn't modelled beyond the spacing-dependent fringe-capacitance term already built into the Cu capacitance model.
- **Room-Temperature Only:** All results assume $T = 300$ K. Temperature-dependent effects on Cu grain-boundary scattering and CNT phonon scattering – both physically sensitive to temperature – aren't swept here.
- **No Delay/PDP/EDP Evaluation Performed:** The Elmore delay, PDP, and EDP formulas are laid out, but a full delay and energy comparison across all node-length combinations is left for future work.

5.5 Future Scope

5.5.1 Modelling Improvements

- **Sensitivity Analysis:** Sweep F_m (0.33 to 1.0, covering unsorted through fully sorted synthesis) and $R_{contact}$ (3.2 to 100 kΩ) to see how much each one shifts the Cu-CNT crossover point.
- **Delay, PDP, and EDP Evaluation:** Build out the current RLC comparison into a full Elmore delay, power-delay product, and energy-delay product evaluation across all node-length combinations, using the framework already laid out in Section 3.5.
- **Coupled-Line Crosstalk:** Add mutual inductance and coupling capacitance between adjacent lines to better reflect a real multi-wire on-chip environment.

5.5.2 Fabrication-Aware Modelling

- **Sorted-Tube Bundles:** Model what improved CNT sorting ($F_m \rightarrow 1.0$) would do – the literature suggests it could cut bundle resistance by roughly $3\times$.
- **Improved Contact Engineering:** Model best-case palladium contact resistance (3.2 kΩ) to establish an upper bound on achievable CNT performance.

5.5.3 Circuit-Level Extension

- **Larger Benchmark Circuits:** Push circuit-level validation past the inverter chain and 4-bit RCA to larger arithmetic and memory-array blocks that better represent real global-tier wiring.
- **Mixed Cu-CNT Hybrid Interconnects:** Look at hybrid schemes that use Cu for local/intermediate tiers and CNT bundles for global tiers, where CNT's length scaling advantage matters most.

5.6 Final Remarks

This project shows that copper's resistivity bottleneck turns severe below 14 nm, while SWCNT bundle interconnects – despite being held back by quantum and contact resistance at short lengths – scale more favourably with both length and technology node. The crossover between the two materials is real and node-dependent, not universal: Cu still wins at 45 nm, but the gap closes sharply by 7 nm. And because CNT's higher capacitance means RC delay doesn't automatically improve just because resistance is comparable, both R and C need to be optimised together for CNT bundles to offer a real advantage. With sorted-tube synthesis and better contact engineering, CNT bundles could become viable for global interconnects beyond 500 µm at 7 nm and below – making them a promising, if contact-resistance-limited, alternative for future VLSI interconnects.

Chapter 6

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